

Privacy, Safety and Wellbeing: Solutions for the Future of AR and VR

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ABSTRACT

As new immersive technology platforms are developed, it's imperative that they're designed responsibly and in a way that protects people's privacy, safety, and wellbeing. Today, companies, as well as AR/VR creators, need to be cognizant of the potential societal impacts of the XR platforms and experiences they're building. This panel discusses:

- Privacy-centric technology best practices from the hardware to application layers.
- Ethical and transparent information collection with HMDs, including the impacts to indigenous and minority cultures.
- Privacy by design in open-source XR frameworks.

The panelists have a deep expertise in these areas spanning industry and academia. Dr. Eakta Jain brings perspective on eye tracking, surveillance, policy and privacy-preserving technical solutions. Liv Erickson shares insights from open-source XR communities and designing privacy-centric XR platforms for the web. Michael Running Wolf intersects academia and large MANGA companies as a lecturer and software engineer respectively, formerly working on Amazon Alexa's privacy and compliance team. His passion is how XR platforms can revive indigenous culture responsibly and ethically. Callie Holderman builds developer platforms for immersive technologies. The panelists discuss XR privacy's ethical considerations and provide recommendations for future research including software design and interoperability.

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1 INTRODUCTION

Society is no stranger to connected, online spaces and the threat mass-surveillance that governments and corporations pose to their

privacy. Anonymized internet forums gave way to instant messenger which led to social media platforms: with each social layer of the internet, a public individual identity arose while individual privacy ebbed. Today's social internet sees rapid proliferation of data-collecting services. AR/VR platforms and mobile devices connect humanity in immersive worlds while also tracking human physiology. This brings new mechanisms for data collection to the market that tabulate identity, emotional state, location and relationships with granularity.

2 DISCUSSION

2.1 HMDs and Personal Data Collection

AR and VR HMDs capture and record sensitive data that divulges emotional state and cognition. This rich and powerful data enables new applications that improve our quality of life, while introducing new security and privacy considerations. Eye tracking systems record individuals' gaze data, which can be used to detect emotional states like arousal and cognitive states like schizophrenia [Wolf et al., 2021]. Who has access to this data and controls it has implications for individuals, especially marginalized communities.

2.2 Consumer Sentiments on Privacy

Current consumer sentiments on privacy suggest consumers do care greatly about their privacy and take means to control their data [Outlaw, 2021]. However, they also tend to underestimate the potential privacy harms that can befall them. Kroger et al. showed that consumer data that was given freely could be used for everything from discrimination to discrediting, to just the banal - targeting people with junk mail [Kröger et al., 2021]. When it comes to the biometric data streams collected by HMDs such as eye gaze, consumers are not well-versed in the data itself, nor the potential inferences and predictions that could be made about them based on it. Plus, passive, ego-centric data collection by augmented reality glasses and other wearable devices has the potential to collect data on bystanders, their routines and their actions at scale while matching behaviors at an individual level using computer vision. These novel data streams will pose a challenge for consumers to manage individually, which means that technologists must consider the potential consequences of collecting and aggregating this data over time.

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3 RECOMMENDATIONS

3.1 Privacy-centric technology best practices

Adhering to privacy-by-design practices and giving individuals control and transparency to privacy settings are two ways to incorporate best practices into software, firmware and hardware design.

One example of privacy-by-design is the gatekeeper model [David-John et al., 2021] for networked XR systems and sensory application programming interfaces (APIs), to control what data is passed through to the next application layer. Data-minimal decisions, such as introducing mechanisms to protect identity during gaze detection, still result in less than 1.5 error in gaze position and prediction while protecting identification 70% of the time (compared to 15% without such mechanisms) [David-John et al., 2021]. Gate-keeping what data is exposed through the API means there is no downstream implication around user-consent of its use.

In cases where user data cannot be obscured within the API design, UI designers play an important role to expose privacy setting defaults. New UI creation is necessary for XR devices because physiological sensors are necessary for device operation; hands-free HMDs require sensory cameras for basic operations. More research is needed in this area to understand when to surface UI and how.

3.2 Privacy by design in open-source XR

Interoperability between devices, digital ecosystems, and real-world systems is symbiotic for commerce, communities and their privacy. Privacy-at-scale requires platform interoperability and open source standards to honor privacy across digital platforms.

OpenXR provides a cross-platform API for devices and applications, WebXR similarly does this for the web. These standards

do not provide a model for implementation. Platforms that adopt such standards can operate within a larger XR ecosystem. If a platform is not adhering to privacy-by-design practices and thus losing end-users, developers can easily bring their content to another platform.

Incorporating standards into a nascent platform can be challenging. More research and support for open frameworks is needed to bolster consumer trust and mass AR/VR adoption.

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